MODULE DESCRIPTOR

M5: Machine Learning

Module Information

Programme	Master of Science (MSc) in Computer Science
Module Code	M5
Module Title	Machine Learning
MQF Level	7
ECTS Credits	10
Total Learning Hours	250 hours (50 contact hours, 160 self-study, 40 assessment)
Sessions	16 sessions x 3 hours
Mode of Delivery	Face-to-Face (Lab-based)
Assessment	Written Exam (60%) | Kaggle Capstone Project (30%) | Presentation (10%)
Pass Mark	50%
Pre-requisites	M3 or M4; Python proficiency; basic statistics

Module Description

The Machine Learning module equips students with comprehensive practical skills in data-driven decision making. Students master the full ML workflow: from data collection and preprocessing through feature engineering, model selection, evaluation, and deployment. The module covers supervised, unsupervised, and reinforcement learning, culminating in a Kaggle-style competition capstone project. Python (scikit-learn, TensorFlow, Keras) is the primary toolset throughout.

Learning Outcomes

Code	Learning Outcome
LO1	Explain and compare supervised, unsupervised, and reinforcement learning paradigms, identifying appropriate algorithms for given problem types.
LO2	Implement data preprocessing pipelines including cleaning, feature engineering, scaling, and encoding using Python (pandas, scikit-learn).
LO3	Build, train, and evaluate supervised learning models (regression and classification) using scikit-learn, applying rigorous cross-validation.
LO4	Apply unsupervised learning techniques (clustering, dimensionality reduction) and interpret results in a domain context.
LO5	Develop reinforcement learning solutions by defining MDPs and implementing Q-learning or policy gradient methods.
LO6	Build and train deep neural networks using TensorFlow/Keras, applying regularisation techniques to prevent overfitting.
LO7	Evaluate model performance using appropriate metrics, diagnose bias-variance issues, and apply hyperparameter optimisation.

LO8	Design and execute an end-to-end machine learning competition pipeline (Kaggle-style), document findings, and present results.

Indicative Content & Session Map

Sessions	Topic Block	Key Content
S1–S2	**ML Foundations & Python Stack**	ML paradigms, Python scientific stack, data wrangling, EDA
S3–S4	**Supervised Learning I: Regression**	Linear/polynomial regression, regularisation, model evaluation
S5–S6	**Supervised Learning II: Classification**	Decision trees, random forests, SVM, ensemble methods
S7–S8	**Model Evaluation & Optimisation**	Cross-validation, hyperparameter tuning, feature importance
S9–S10	**Unsupervised Learning**	K-means, hierarchical, DBSCAN, PCA, t-SNE, autoencoders
S11–S12	**Reinforcement Learning**	MDP, Q-learning, OpenAI Gym, policy gradient introduction
S13–S14	**Deep Learning with TensorFlow/Keras**	MLP, CNN, RNN basics, transfer learning, model deployment
S15	**Advanced Topics: Ethics & Fairness**	Bias in ML, fairness metrics, responsible ML, explainability
S16	**Capstone Presentations & Exam Prep**	Kaggle challenge debrief, presentations, module synthesis

Assessment

Written Exam (60%): 3-hour end-of-module exam. Questions cover theoretical understanding of ML algorithms, mathematical foundations, and interpretation of results.

Kaggle Capstone Project (30%): Students compete in a class-internal Kaggle-style challenge (provided dataset). Graded on final model performance (leaderboard) plus a technical report documenting methodology, experiments, and findings.

Presentation (10%): 10-minute presentation of capstone methodology and results to the class.

Reading List

Type	Reference
Core	Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.
Core	Goodfellow, I., Bengio, Y. & Courville, A. (2016). Deep Learning. MIT Press.
Core	Hastie, T., Tibshirani, R. & Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.
Supplementary	Provost, F. & Fawcett, T. (2013). Data Science for Business. O'Reilly Media.
Supplementary	Ng, A. (2017). Machine Learning Yearning. Available at: http://www.mlyearning.org/